

DIGITAL FILTERS FOR CURRENT AND VOLTAGE ACQUISITION IN MIG/MAG WELDING PROCESSES

1st Christian Griesang Barbosa
Digital Processing Class (PPGEE)
Federal University of Santa Maria (UFSM)
Santa Maria, RS, Brazil
chtn_@hotmail.com

Abstract—This work demonstrates the importance of using digital filters in the accurate acquisition of welding parameters, such as current and voltage, before they are used by the control system. By applying low-pass FIR filters to real monitored signals, we were able to significantly reduce the noise present in the data. A comparison between the filtered signals and the original signals will be presented, highlighting the effectiveness of digital filters in improving data quality and measurement accuracy.

Index Terms—Welding, Current, Voltage, Filter, Hamming

I. INTRODUCTION

The process commonly done on discrete signals is the operation known as filtering. Filtering a signal means selecting from that signal which frequencies are desired, and discarding the remaining [1]. This technique is fundamental in signal processing and can be carried out using different methods, such as low-pass, high-pass, band-pass and band-reject filters, each suitable for different applications and needs.

Welding parameters need to be measured, monitored and optimized to achieve desired results. Therefore, the choice of ideal welding parameters must be as precise as possible to guarantee maximum weld quality. To achieve this, it is necessary to establish an appropriate model that relates welding parameters to quality attributes. Furthermore, the implementation of advanced monitoring and control techniques can help to continuously adjust these parameters, ensuring consistent, high-quality welding [2].

Correctly filtering the current and voltage reading in the MIG/MAG process before the signal is used by the control system is essential to guarantee the precision, stability and quality of the process, in addition to protecting the equipment against damage. The elimination of unwanted noise and fluctuations allows for more efficient adjustments of welding parameters, improving productivity and control on the part of the welder.

Furthermore, an accurate and stable measurement of welding parameters contributes to the consistency of the process, ensuring the quality and stability of the arc, as the presence of noise can lead to aggressive control, causing delays in response or even extinction of the arc.

II. MIG/MAG SHORT CIRCUIT WELDING

Depending on the desired welding current range for the MIG/MAG welding process, the metal transfer can change, being possible to occur in one of three main modes, namely short-circuit, globular or spray transfer, chosen appropriately with based on the application. These modes are defined by the voltage and current settings of the welding source, adjusted by the operator, where the source control system must be capable of working in each of the modes.

The output current and voltage waveforms for short-circuit transfer mode are illustrated in Figure 1, representing how the consecutive short circuit process occurs during welding.

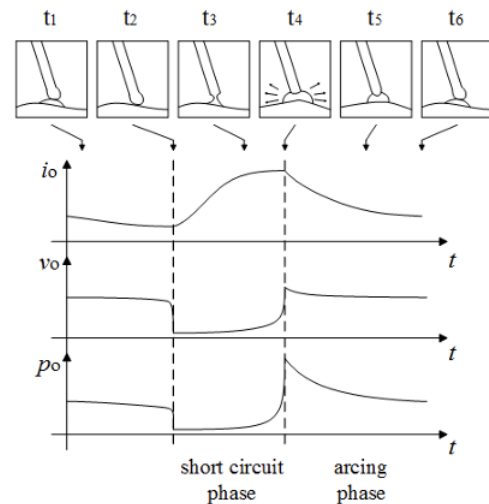


Fig. 1. Waveforms for Output Current and Voltage for MIG/MAG Welding in Short Circuit Transfer mode, Fonte [3].

As shown in Figure 1, at time t_1 , the arc is decreasing. Then at t_2 , when the molten metal at the end of the wire touches the welding puddle, a short circuit occurs, causing the voltage to be reduced and the current to begin to increase. The current increases to a high level at t_3 , which can generate a magnetic force to adjust the connection between the wire and the puddle.

When the droplet detaches between the wire and the welding puddle at t_4 , the arc will restart and the voltage will suddenly increase as the current will begin to decrease. After that, at t_5 , the arc continuously generates heat to melt the end of the wire and form a molten droplet for the next short-circuit metal transfer. Therefore, at t_6 , when the wire approaches the welding pool, the arc begins to decrease and the process repeats.

III. DIGITAL FILTERS

A wide range of power electronics control systems use signals that are DC by definition, being exposed to various noise sources that contaminate them. Therefore, filtration becomes necessary to maintain system performance. Signal set point changes must be captured with the shortest possible delay, making the time-domain characteristics of the applied filters generally more important than the frequency-domain characteristics. Therefore, the step response of the filter is of paramount importance [4].

Based on the filtering algorithm applied, digital filters are typically classified into two types: Finite Impulse Response (FIR) filters, known as non-recursive as the current output is calculated only from the current and previous input values ; and Infinite Impulse Response (IIR) filters, characterized as recursive, as they use both previous input and output values [5]. This distinction is fundamental in the choice and design of the filter, directly influencing the stability and frequency response of the filtered system.

Finite-duration impulse response (FIR) filters are the most popular for implementation in software. Its operation is governed by linear difference equations with constant coefficients, non-recursive in nature. This filter structure can be used to implement almost any type of frequency response digitally, offering advantages such as guaranteed stability and a linear phase, which is crucial in many signal processing applications [6].

In carrying out this work, we chose to choose FIR filters for filtering welding current and voltage due to their stability, direct control over the phase response and ease of design, where these characteristics make FIR filters a robust and reliable.

A. HAMMING WINDOW

In signal processing, a window function is a mathematical function with a value of zero outside some chosen range. It represents a statistical tool that allows users to observe a small region of a signal, usually a flow, with minimal leakage from any other part of the signal [7].

Characterized by good frequency resolution, reduced spectral leakage and acceptable noise performance, the Hamming window, illustrated in Figure 2, is used in communication and computing to smooth data before applying Fourier analysis [7].

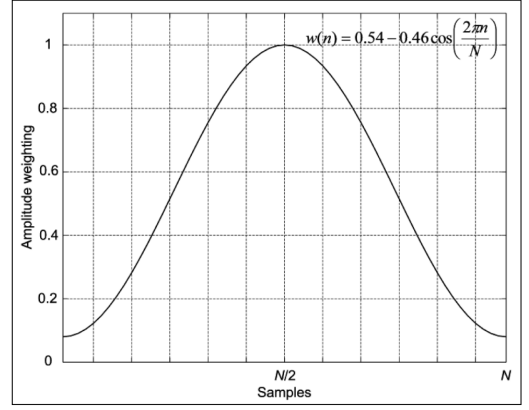


Fig. 2. Hamming Window, Fonte [7].

IV. DESIGN OF FIR FILTERS FOR WELDING CURRENT AND VOLTAGE

To carry out the design of the filters, we will separate it into two stages, with the determination of the filter for the welding current initially, and then for the welding voltage.

A. Welding Current Filter

When designing the welding current filter, the following requirements will be considered as specifications for the project:

- Cutoff Frequency ($f_{c,current}$): 4 kHz
- Sampling Rate (f_s): 80 kHz
- Pass Frequency (f_{pass}): 3 kHz
- Attenuation (A): 60 dB

Calculating the normalized frequencies, where we divide the pass and cut frequencies by half the sampling frequency, we have:

$$W_{pass, current} = \frac{3000}{f_s/2} = \frac{3000}{80000/2} = 0.075$$

$$W_{stop, current} = \frac{4000}{f_s/2} = \frac{4000}{80000/2} = 0.1$$

To determine the filter order, we use the kaiserord function and the normalized frequency (W_n):

$$[N_{current}, W_{n,current}] = \text{kaiserord}([W_{pass, current}, W_{stop, current}], [1, 0], [0.01, 0.001])$$

Finally, we design the FIR filter using the Hamming window. For this, the 'fir1' function was used, which receives the filter order (N), the normalized cutoff frequency (W_n), the filter type ('low' for low-pass) and the Hamming window.

$$\text{hamming_window}_{current} = \text{hamming}(N_{current} + 1)$$

$$b_{current} = \text{fir1}(N_{current}, W_{n,current}, 'low', \text{hamming_window}_{current})$$

The impulse response is demonstrated in Figure 3, where we can observe the stability characteristic of the filter when returning to zero after a finite number of samples.

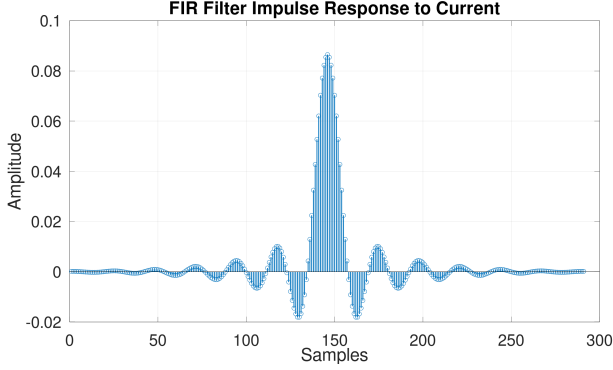


Fig. 3. Impulse Response for FIR Filter for Current, Fonte [Autor].

The frequency response for the scaled filter is illustrated in Figure 4. It is observed that up to a frequency of 3 kHz, the filter has unity gain and then the current signal attenuates.

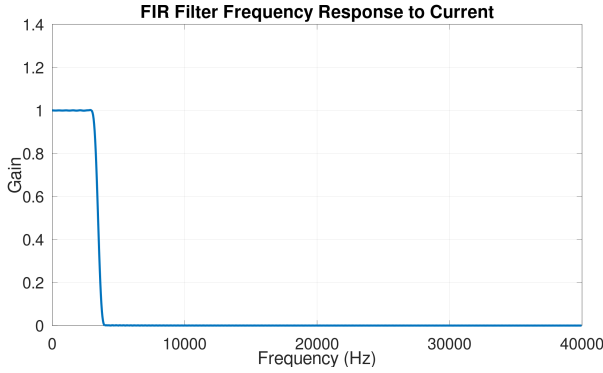


Fig. 4. Frequency Response for FIR Filter for Current, Fonte [Autor].

The Hamming window is illustrated in Figure 5, displaying a function that varies smoothly and approaches zero at the ends.

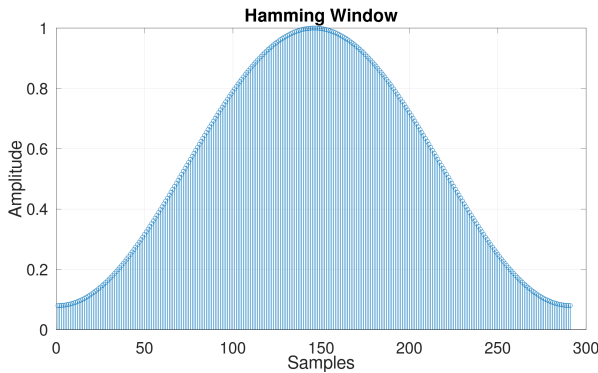


Fig. 5. Hamming Window for Current, Fonte [Autor].

B. Welding Voltage Filter

When designing the filter for welding voltage, the following requirements will be considered as specifications for the project:

- Cutoff Frequency ($f_{c,current}$): 8 kHz
- Sampling Rate (f_s): 80 kHz
- Pass Frequency (f_{pass}): 7 kHz
- Attenuation (A): 60 dB

Calculating the normalized frequencies, where we divide the pass and cut frequencies by half the sampling frequency, we have:

$$W_{pass, voltage} = \frac{6000}{f_s/2} = \frac{6000}{80000/2} = 0.15$$

$$W_{stop, voltage} = \frac{8000}{f_s/2} = \frac{8000}{80000/2} = 0.2$$

To determine the filter order, we use the kaiserord function and the normalized frequency (W_n):

$$[N_{voltage}, W_{n,voltage}] = \text{kaiserord}([W_{pass, voltage}, W_{stop, voltage}], [1, 0], [0.01, 0.001])$$

Finally, we design the FIR filter using the Hamming window. For this, the 'fir1' function was used, which receives the filter order (N), the normalized cutoff frequency (W_n), the filter type ('low' for low-pass) and the Hamming window.

$$\text{hamming_window}_{voltage} = \text{hamming}(N_{voltage} + 1)$$

$$b_{voltage} = \text{fir1}(N_{voltage}, W_{n,voltage}, 'low', \text{hamming_window}_{voltage})$$

The impulse response is demonstrated in Figure 6, where we can observe the stability characteristic of the filter when returning to zero after a finite number of samples.

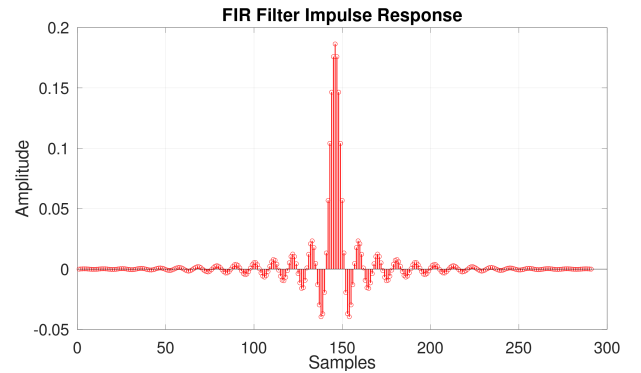


Fig. 6. Impulse Response for FIR Filter for Voltage, Fonte [Autor].

The frequency response for the scaled filter is illustrated in Figure 7. It is observed that up to a frequency of 7 kHz, the filter has unity gain and after that the voltage signal attenuates.

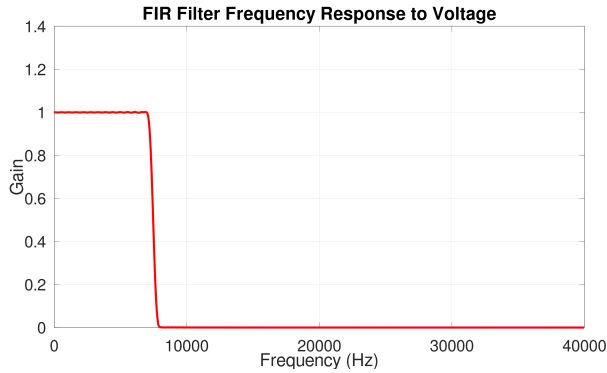


Fig. 7. Frequency Response for FIR Filter to Voltage, Fonte [Autor].

The Hamming window is illustrated in Figure 8, displaying a function that varies smoothly and approaches zero at the ends.

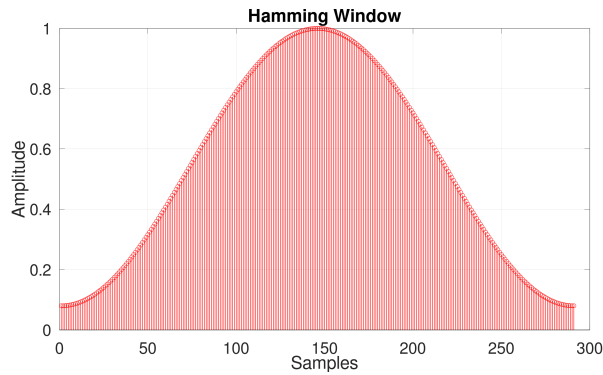


Fig. 8. Hamming Window for Voltage, Fonte [Autor].

V. FILTER APPLICATION

To verify the efficiency of the filter, measurements of current and voltage signals were carried out. Using an oscilloscope, the waveforms were saved in a .csv file, this way, it is possible to import the measured data into the OCTAVE GNU software. Below they are demonstrated in Figure 9. the current and voltage signals, in blue and red, respectively, before applying the filters. These graphs provide a basis for comparing raw and filtered signals, allowing you to evaluate the effectiveness of filtering in reducing noise and preserving signal characteristics.

With the filter applied to the measured signals, we obtain a clean signal, as illustrated in Figure 10, where the current and voltage signals are demonstrated, in blue and red, respectively, observing the effective application of the FIR filter with the Hamming window in each signal, highlighting the attenuation of unwanted noise and the preservation of signal characteristics without suffering attenuation.

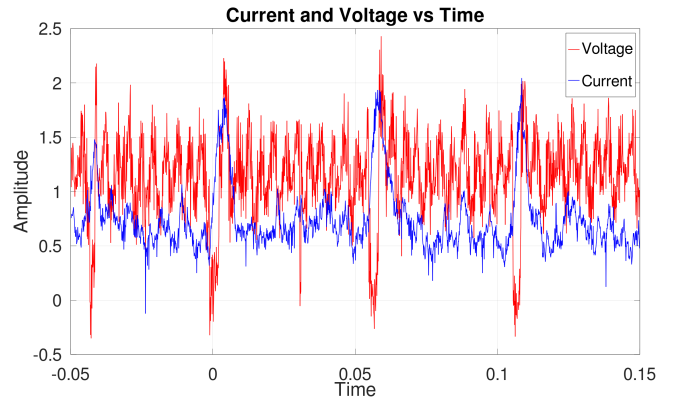


Fig. 9. Welding Current and Voltage without Filter, Fonte [Autor].

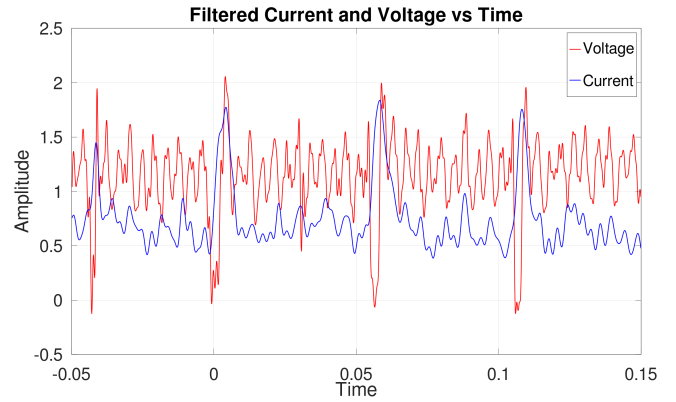


Fig. 10. Welding Current and Voltage with Filtering, Fonte [Autor].

VI. CONCLUSION

The application of the FIR filter using the Hamming window to current and voltage signals resulted in a significant improvement in data quality. The signals showed effective attenuation of high frequency components, indicating that unwanted noise was minimized without compromising the relevant information of the original signals acquired through measurement with the oscilloscope. The reduction of noise peaks is clearly observable in the filtered current and voltage graphs.

Furthermore, the general shape of the current and voltage signals was preserved after filtering, demonstrating that the filters were well designed to maintain the characteristics of the signals. This preservation is crucial to ensure that important information is maintained during the filtering process and so the control system can act more accurately.

REFERENCES

- [1] BENTES, Ana Karina Monteiro; DE SOUSA JÚNIOR, Moacir Carvalho. Filtro Passa-Baixas (FIR): Aplicação na Filtragem de um Sinal de Áudio utilizando a Janela de Hamming.
- [2] GUPTA, Kapil; DAVIM, J. Paulo (Ed.). Advanced welding and deformation. Elsevier, 2021.
- [3] WU, Mingfei; FLYNN, David. An advanced gas metal arc welding machine design for low spatter welding. In: 2018 IEEE 27th International Symposium on Industrial Electronics (ISIE). IEEE, 2018. p. 335-340.

- [4] LUUKKO, Julius; RAUMA, Kimmo. Open-loop adaptive filter for power electronics applications. IEEE Transactions on Industrial Electronics, v. 55, n. 2, p. 910-917, 2008.
- [5] SHEN, Hao. Seam position detection in pulsed gas metal arc welding. 2003.
- [6] MYNUDDIN, Mohammed et al. Designing a low-pass fir digital filter by using hamming window and Blackman window technique. Science Journal of Circuits, Systems and Signal Processing, v. 10, n. 1, p. 9-13, 2015.
- [7] BOJKOVIC, Zoran S.; BAKMAZ, Bojan M.; BAKMAZ, Miodrag R. Hamming window to the digital world. Proceedings of the IEEE, v. 105, n. 6, p. 1185-1190, 2017.